

2015 HSC Mathematics Extension 2 Marking Guidelines

Section I

IMPORTANT:

This resource was developed to support a previous version of the syllabus.
It may contain content that differs from the current syllabus.

Multiple-choice Answer Key

Question	Answer
1	D
2	A
3	A
4	C
5	B
6	C
7	A
8	B
9	D
10	C

Section II

Question 11 (a)

Criteria	Marks
• Provides correct solution	2
• Attempts to use conjugate of denominator, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \frac{4+3i}{2-i} &= \frac{(4+3i)}{(2-i)} \times \frac{(2+i)}{(2+i)} = \frac{8+6i+4i-3}{4+1} \\
 &= \frac{5+10i}{5} \\
 &= 1+2i
 \end{aligned}$$

Question 11 (b) (i)

Criteria	Marks
• Provides correct answer	1

Sample answer:

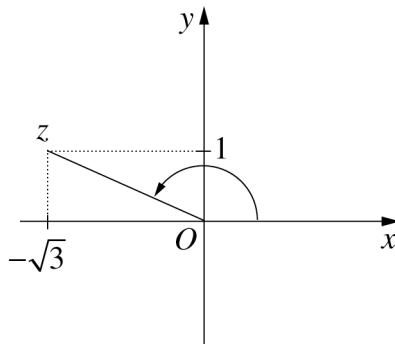
$$\begin{aligned}
 |z| &= |-\sqrt{3}+i| \\
 &= \sqrt{(-\sqrt{3})^2+1} \\
 &= \sqrt{3+1} \\
 &= 2
 \end{aligned}$$

Question 11 (b) (ii)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\begin{aligned}
 \arg z &= \pi + \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right) \\
 &= \pi - \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) \\
 &= \frac{5\pi}{6}
 \end{aligned}$$

**Question 11 (b) (iii)**

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\begin{aligned}
 \arg\left(\frac{z}{w}\right) &= \arg z - \arg w \\
 &= \frac{5\pi}{6} - \frac{\pi}{7} \\
 &= \frac{29\pi}{42}
 \end{aligned}$$

Question 11 (c)

Criteria	Marks
• Provides correct solution	2
• Finds A, or equivalent merit	1

Sample answer:

$$\frac{1}{x(x^2 + 2)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 2}$$

$$1 = A(x^2 + 2) + (Bx + C)x$$

$$1 = Ax^2 + 2A + Bx^2 + Cx$$

Equating constant terms,

$$1 = 2A$$

$$A = \frac{1}{2}$$

Equating coefficients of x^2 ,

$$0 = A + B$$

$$0 = \frac{1}{2} + B$$

$$B = -\frac{1}{2}$$

Equating coefficients of x ,

$$C = 0$$

Question 11 (d)

Criteria	Marks
• Provides correct sketch	2
• Finds eccentricity, or equivalent merit	1

*Sample answer:*For an ellipse $b^2 = a^2(1 - e^2)$

$$\Rightarrow 16 = 25(1 - e^2)$$

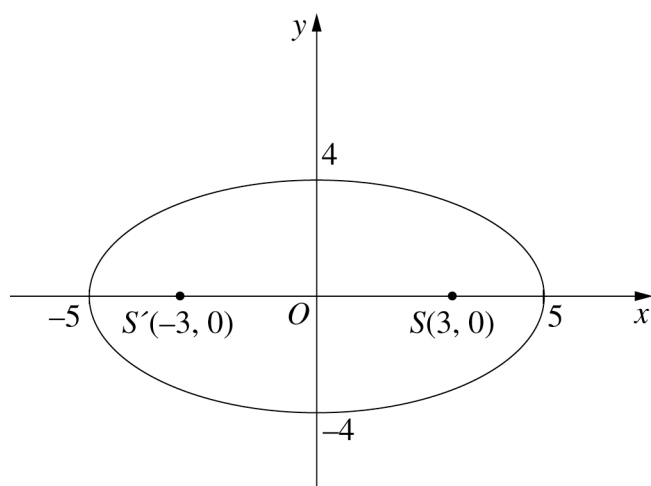
$$\frac{16}{25} = 1 - e^2$$

$$e^2 = 1 - \frac{16}{25}$$

$$e = \frac{3}{5}$$

$$\therefore \text{Foci } (\pm ae, 0) = \left(\pm 5 \times \frac{3}{5}, 0 \right)$$

$$= (\pm 3, 0)$$



Question 11 (e)

Criteria	Marks
• Provides correct solution	3
• Correctly differentiates, or equivalent merit	2
• Uses the chain rule, or equivalent merit	1

Sample answer:

Differentiating with respect to x

$$1 + 2xy^3 + x^2 3y^2 \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{-1 - 2xy^3}{x^2 3y^2}$$

At $(2, -1)$

$$\begin{aligned} \frac{dy}{dx} &= \frac{-1 - 2 \times 2 \times (-1)^3}{2^2 \times 3 \times (-1)^2} \\ &= \frac{1}{4} \end{aligned}$$

Question 11 (f) (i)

Criteria	Marks
• Provides correct solution	2
• Expresses $\cot \theta$ or $\operatorname{cosec} \theta$ in terms of $\tan \frac{\theta}{2}$, or equivalent merit	1

Sample answer:

$$LHS = \cot \theta + \operatorname{cosec} \theta \quad \text{Let } t = \tan \frac{\theta}{2}$$

$$= \frac{1-t^2}{2t} + \frac{1+t^2}{2t}$$

$$= \frac{2}{2t}$$

$$= \frac{1}{t}$$

$$= \cot \left(\frac{\theta}{2} \right)$$

$$= RHS$$

Question 11 (f) (ii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

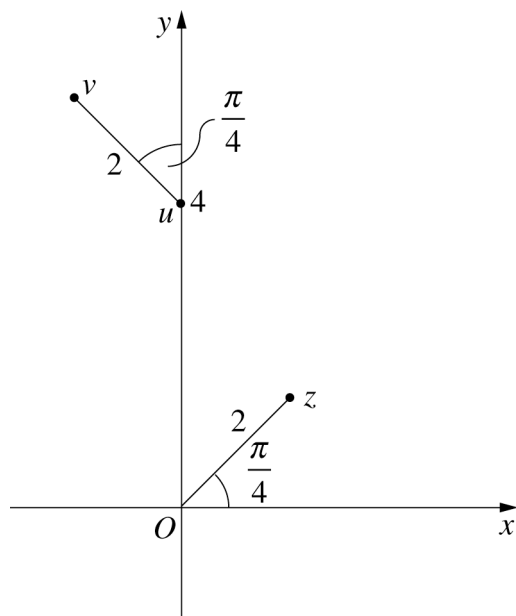
$$\int (\cot \theta + \operatorname{cosec} \theta) d\theta$$

$$= \int \cot \frac{\theta}{2} d\theta$$

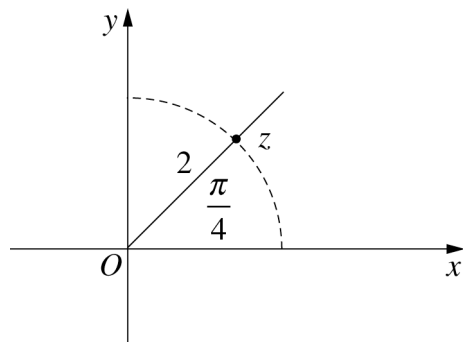
$$= \int \frac{\cos \frac{\theta}{2}}{\sin \frac{\theta}{2}} d\theta$$

$$= \frac{1}{\frac{1}{2}} \ln \left| \sin \frac{\theta}{2} \right| + c$$

$$= 2 \ln \left| \sin \frac{\theta}{2} \right| + c$$

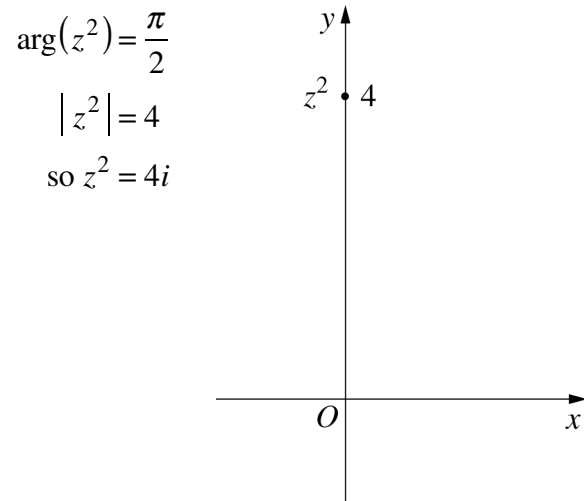
Question 12 (a) (i), (ii), (iii)**Question 12 (a) (i)**

Criteria	Marks
• Provides correct solution	1

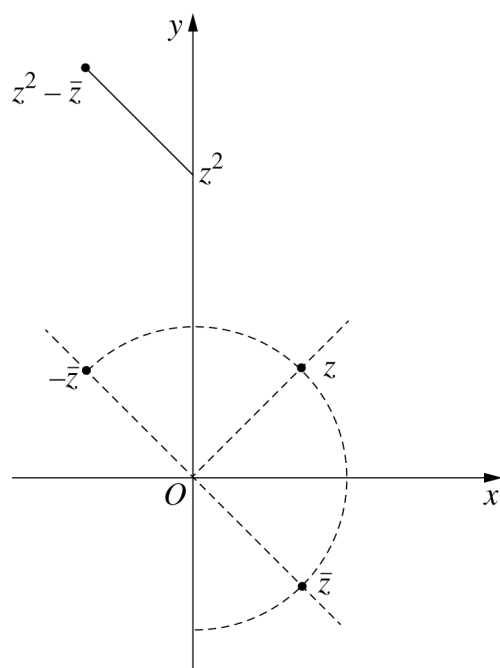
Sample answer:

Question 12 (a) (ii)

Criteria	Marks
• Provides correct solution	1

Sample answer:**Question 12 (a) (iii)**

Criteria	Marks
• Provides correct solution	1

Sample answer:

Question 12 (b) (i)

Criteria	Marks
• Provides correct solution	3
• Finds a , or equivalent merit	2
• Recognises that the conjugates are also roots, or equivalent merit	1

Sample answer:

The roots are $a \pm ib$ and $a \pm 2ib$

$\therefore$ The sum of the roots is

$$(a + 2ib) + (a - 2ib) + (a + ib) + (a - ib) = 4$$

$$\Rightarrow 4a = 4$$

$$a = 1$$

The product of the roots is

$$(a + 2ib)(a - 2ib)(a + ib)(a - ib) = 10$$

$$\Rightarrow (a^2 + 4b^2)(a^2 + b^2) = 10$$

But $a = 1$,

$$\therefore (1 + 4b^2)(1 + b^2) = 10$$

$$4b^4 + 5b^2 + 1 - 10 = 0$$

$$4b^4 + 5b^2 - 9 = 0$$

$$(4b^2 + 9)(b^2 - 1) = 0$$

$$b^2 = 1$$

$$b = \pm 1$$

$\therefore$ Roots are $1 \pm i$, $1 \pm 2i$

Question 12 (b) (ii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

The roots are $x = 1 \pm i$ and $x = 1 \pm 2i$

Polynomials with real coefficients have roots that occur in complex conjugate pairs.

$\therefore$ Using the sum and product of the roots, the required quadratics are:

$$\begin{aligned} x^2 - (1+i+1-i)x + (1+i)(1-i) & \text{ and } x^2 - (1+2i+1-2i)x + (1+2i)(1-2i) \\ = x^2 - 2x + 2 & \qquad \qquad \qquad = x^2 - 2x + 5 \end{aligned}$$

Question 12 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Attempts a division of polynomials, or equivalent merit	1

Sample answer:

$$\frac{(x-2)(x-5)}{(x-1)} = \frac{x^2 - 7x + 10}{x-1}$$

Using long division,

$$\begin{array}{r} x-6 \\ x-1 \overline{) x^2 - 7x + 10} \\ \underline{x^2 - x} \\ -6x + 10 \\ \underline{-6x + 6} \\ 4 \end{array}$$

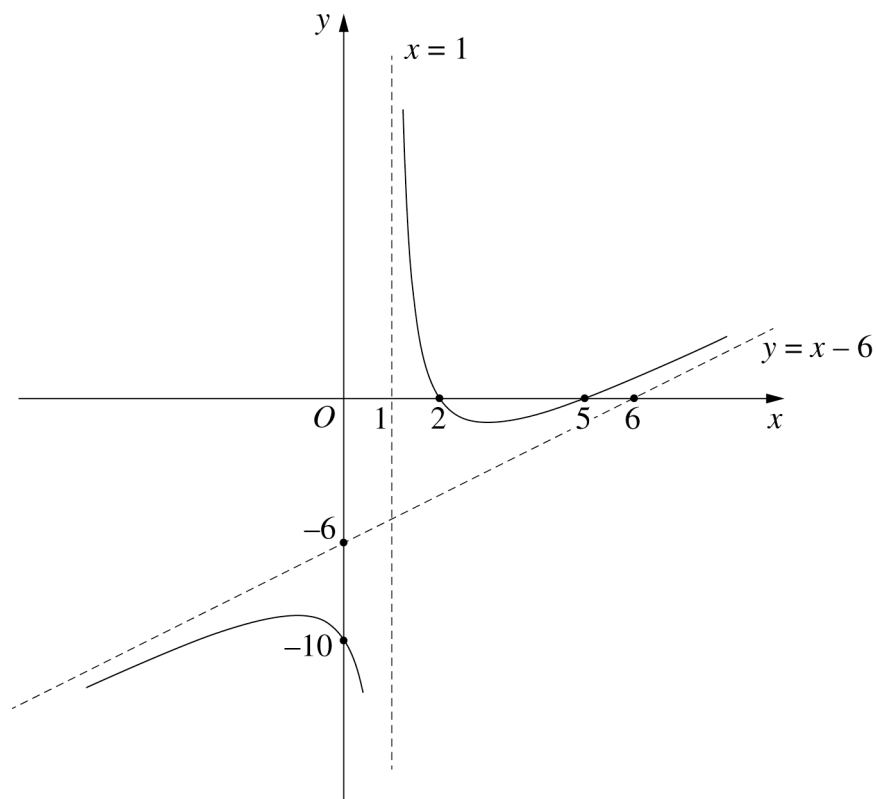
$$\therefore \frac{(x-2)(x-5)}{x-1} = x - 6 + \frac{4}{x-1}$$

where $m = 1$, $b = -6$, $a = 4$ $\therefore$ The oblique asymptote is $y = x - 6$

Question 12 (c) (ii)

Criteria	Marks
• Provides correct sketch	2
• Identifies any TWO intercepts and the vertical asymptote, or equivalent merit	1

Sample answer:



Question 12 (d)

Criteria	Marks
• Provides correct solution	4
• Attempts to evaluate the correct integral using a suitable method, or equivalent merit	3
• Provides correct integral expression for the volume, or equivalent merit	2
• Recognises the radius is $3 - x$, or equivalent merit	1

Sample answer:

Radius of a typical shell is $3 - x$

$$\therefore V = 2\pi \int_0^3 (3-x)\sqrt{x+1} dx$$

$$\text{Let } u^2 = x+1$$

$$x = u^2 - 1$$

$$\frac{dx}{du} = 2u$$

$$x = 0, \quad u = 1$$

$$x = 3 \quad u = 2$$

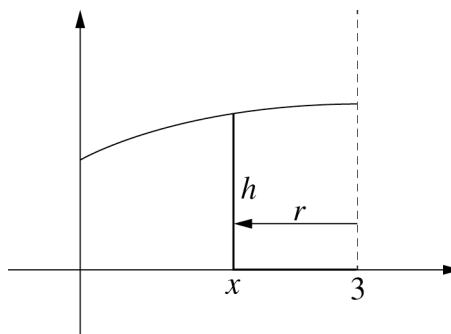
$$\therefore V = 2\pi \int_1^2 (4-u^2)u \cdot 2u du$$

$$= 4\pi \int_1^2 (4u^2 - u^4) du$$

$$= 4\pi \left[\frac{4u^3}{3} - \frac{u^5}{5} \right]_1^2$$

$$= 4\pi \left(\frac{32}{3} - \frac{32}{5} - \left(\frac{4}{3} - \frac{1}{5} \right) \right)$$

$$= \frac{188\pi}{15} \text{ units}^3$$



area of a typical shell = $2\pi rh$

Question 13 (a) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:At Q , $x = a \tan \theta$, $y = b \sec \theta$, so substituting into H_2

$$\begin{aligned}
 LHS &= \frac{(a \tan \theta)^2}{a^2} - \frac{(b \sec \theta)^2}{b^2} \\
 &= \frac{a^2 \tan^2 \theta}{a^2} - \frac{b^2 \sec^2 \theta}{b^2} \\
 &= \frac{a^2 b^2 (\tan^2 \theta - \sec^2 \theta)}{a^2 b^2} \\
 &= \frac{a^2 b^2 (-1)}{a^2 b^2} \quad (\text{since } \tan^2 \theta + 1 = \sec^2 \theta) \\
 &= -1 \\
 &= RHS
 \end{aligned}$$

Question 13 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Finds the slope of PQ , or equivalent merit	1

Sample answer:

The gradient of PQ is $\frac{b \sec \theta - b \tan \theta}{a \tan \theta - a \sec \theta}$

$$\begin{aligned}
 &= \frac{-b(\tan \theta - \sec \theta)}{a(\tan \theta - \sec \theta)} \\
 &= \frac{-b}{a}
 \end{aligned}$$

 $\therefore$ equation of PQ is

$$\begin{aligned}
 y - b \tan \theta &= \frac{-b}{a}(x - a \sec \theta) \\
 \Rightarrow ay - ab \tan \theta &= -bx + ab \sec \theta \\
 \Rightarrow bx + ay &= ab(\tan \theta + \sec \theta)
 \end{aligned}$$

Question 13 (a) (iii)

Criteria	Marks
• Provides correct proof	3
• Finds a correct expression for the area, or equivalent merit	2
• Finds a correct expression for one relevant distance, or equivalent merit	1

Sample answer:

$$PQ^2 = a^2(\tan \theta - \sec \theta)^2 + b^2(\sec \theta - \tan \theta)^2$$

$$= (a^2 + b^2)(\tan \theta - \sec \theta)^2$$

$$\therefore PQ = \sqrt{a^2 + b^2} |\tan \theta - \sec \theta|$$

$$= \sqrt{a^2 + b^2} (\sec \theta - \tan \theta)$$

Since for $0 \leq \theta \leq \frac{\pi}{2}$, $\sec \theta \geq \tan \theta$

line PQ : $bx + ay - ab(\tan \theta + \sec \theta) = 0$

Perpendicular distance from O to PQ is

$$= \frac{|b \times 0 + a \times 0 - ab(\tan \theta + \sec \theta)|}{\sqrt{a^2 + b^2}}$$

$$= \frac{ab(\tan \theta + \sec \theta)}{\sqrt{a^2 + b^2}}$$

Now, area of $\triangle OPQ$ is

$$\frac{1}{2} \times \sqrt{a^2 + b^2} (\sec \theta - \tan \theta) \times \frac{ab(\tan \theta + \sec \theta)}{\sqrt{a^2 + b^2}}$$

$$= \frac{ab}{2} (\sec^2 \theta - \tan^2 \theta)$$

$$= \frac{ab}{2} (1) \text{ which is independent of } \theta$$

Question 13 (b) (i)

Criteria	Marks
• Provides correct solution	1

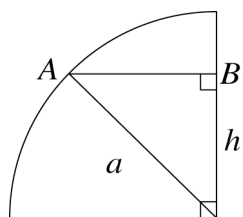
Sample answer:

The cylinder is bounded by a circle $x^2 + y^2 = a^2$

$\therefore$ when $y = h$, $x = \sqrt{a^2 - h^2}$

Or

Using the right hand quarter cylinder



Using Pythagoras' Theorem, $AB = \sqrt{a^2 - h^2}$

Question 13 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Finds correct integral expression for the volume, or equivalent merit	1

Sample answer:

Area of the slice $ABCD = a^2 - h^2$

Hence, the volume is given by

$$V = \int_0^a (a^2 - h^2) dh$$

$$= \left[a^2 h - \frac{h^3}{3} \right]_0^a$$

$$= \frac{2a^3}{3} \text{ units}^3$$

Question 13 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Correctly finds $\frac{dS}{dt}$ in terms of $\frac{dr}{dt}$, or equivalent merit	1

Sample answer:

$$S = 4\pi r^2 \text{ and } \frac{dS}{dt} = \left(\frac{4\pi}{3}\right)^{\frac{1}{3}} \text{ and } \frac{dS}{dt} = \frac{dS}{dr} \cdot \frac{dr}{dt}$$

$$\therefore \frac{dS}{dt} = 8\pi r \frac{dr}{dt}$$

$$\text{so } \left(\frac{4\pi}{3}\right)^{\frac{1}{3}} = 8\pi r \frac{dr}{dt}$$

$$\text{giving } \frac{dr}{dt} = \frac{1}{8\pi r} \left(\frac{4\pi}{3}\right)^{\frac{1}{3}}$$

Question 13 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Correctly finds $\frac{dV}{dt}$ in terms of $\frac{dr}{dt}$, or equivalent merit	1

Sample answer:

$$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$$

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$

$$= 4\pi r^2 \cdot \frac{1}{8\pi r} \left(\frac{4\pi}{3}\right)^{\frac{1}{3}} \text{ by (i)}$$

$$= \frac{1}{2} r \left(\frac{4\pi}{3}\right)^{\frac{1}{3}}$$

$$= \frac{1}{2} \left(r^3 \cdot \frac{4\pi}{3}\right)^{\frac{1}{3}}$$

$$\frac{dV}{dt} = \frac{1}{2} V^{\frac{1}{3}}$$

Question 13 (c) (iii)

Criteria	Marks
• Provides correct solution	2
• Finds correct primitive for t in terms of V , or equivalent merit	1

Sample answer:

$$\text{As } \frac{dV}{dt} = \frac{1}{2} V^{\frac{1}{3}}$$

$$\text{so } \int \frac{dV}{V^{\frac{1}{3}}} = \frac{1}{2} \int dt$$

$$\int_{8\,000}^{64\,000} V^{-\frac{1}{3}} dV = \frac{1}{2} \int_0^t dt$$

$$\therefore \frac{1}{2}t = \left[\frac{3}{2} V^{\frac{2}{3}} \right]_{8\,000}^{64\,000}$$

$$t = 3(40^2 - 20^2)$$

$$\therefore \text{time} = 3\,600 \text{ seconds}$$

$$(\text{= 1 hour})$$

Question 14 (a) (i)

Criteria	Marks
• Provides correct solution	2
• Attempts to differentiate and obtains an expression involving $\cos^2 \theta$, or equivalent merit	1

Sample answer:

$$y = \sin^{n-1} \theta \cos \theta$$

$$y' = u'v + v'u \quad \text{Let } u = \sin^{n-1} \theta \quad v = \cos \theta$$

$$u' = (n-1)\sin^{n-2} \theta \cos \theta \quad v' = -\sin \theta$$

$$y' = (n-1)\sin^{n-2} \theta \cos^2 \theta + \sin^{n-1} \theta (-\sin \theta)$$

$$= (n-1)\sin^{n-2} \theta (1 - \sin^2 \theta) - \sin^n \theta$$

$$= (n-1)\sin^{n-2} \theta - (n-1)\sin^n \theta - \sin^n \theta$$

$$= (n-1)\sin^{n-2} \theta - n\sin^n \theta$$

Question 14 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Attempts to evaluate the integral of the expression from part (i), or equivalent merit	1

Sample answer:

$$y' = \frac{dy}{d\theta} = (n-1)\sin^{n-2}\theta - n\sin^n\theta$$

$$\int_0^{\frac{\pi}{2}} dy = \int_0^{\frac{\pi}{2}} (n-1)\sin^{n-2}\theta - n\sin^n\theta d\theta$$

$$\left[y \right]_0^{\frac{\pi}{2}} = (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2}\theta d\theta - n \int_0^{\frac{\pi}{2}} \sin^n\theta d\theta$$

$$\left[\sin^{n-1}\theta \cos\theta \right]_0^{\frac{\pi}{2}} = (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2}\theta d\theta - n \int_0^{\frac{\pi}{2}} \sin^n\theta d\theta$$

$$0 = (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2}\theta d\theta - n \int_0^{\frac{\pi}{2}} \sin^n\theta d\theta$$

$$n \int_0^{\frac{\pi}{2}} \sin^n\theta d\theta = (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2}\theta d\theta$$

$$\int_0^{\frac{\pi}{2}} \sin^n\theta d\theta = \left(\frac{n-1}{n} \right) \int_0^{\frac{\pi}{2}} \sin^{n-2}\theta d\theta$$

Question 14 (a) (iii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$\begin{aligned}
 \int_0^{\frac{\pi}{2}} \sin^4 \theta d\theta &= \frac{4-1}{4} \int_0^{\frac{\pi}{2}} \sin^2 \theta d\theta \\
 &= \frac{3}{4} \left[\left(\frac{2-1}{2} \right) \int_0^{\frac{\pi}{2}} d\theta \right] \\
 &= \frac{3}{4} \times \frac{1}{2} \left[\theta \right]_0^{\frac{\pi}{2}} \\
 &= \frac{3}{8} \left(\frac{\pi}{2} - 0 \right) \\
 &= \frac{3\pi}{16}
 \end{aligned}$$

Question 14 (b) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$\begin{aligned}
 \alpha + \beta + \gamma &= -\frac{b}{a} = 0 \\
 \alpha\beta + \alpha\gamma + \beta\gamma &= \frac{c}{a} = -p \\
 (\alpha + \beta + \gamma)^2 &= 0 \\
 \text{ie } \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \alpha\gamma + \beta\gamma) &= 0 \\
 16 - 2p &= 0 \\
 \therefore p &= 8
 \end{aligned}$$

Question 14 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Obtains a correct equation involving q and at least one of α^3 , β^3 , γ^3 , or equivalent merit	1

Sample answer: α is a root so

$$\alpha^3 = p\alpha - q \quad (1)$$

similarly $\beta^3 = p\beta - q \quad (2)$

$$\gamma^3 = p\gamma - q \quad (3)$$

adding (1) + (2) + (3)

$$\alpha^3 + \beta^3 + \gamma^3 = p(\alpha + \beta + \gamma) - 3q$$

$$-9 = p \times 0 - 3q$$

$$\therefore q = 3$$

Question 14 (b) (iii)

Criteria	Marks
• Provides correct solution	2
• Obtains an equation involving $\alpha^4 + \beta^4 + \gamma^4$, or equivalent merit	1

Sample answer:

Since $\alpha^3 = p\alpha - q$

so $\alpha^4 = p\alpha^2 - q\alpha$

similarly $\beta^4 = p\beta^2 - q\beta$

$$\gamma^4 = p\gamma^2 - q\gamma$$

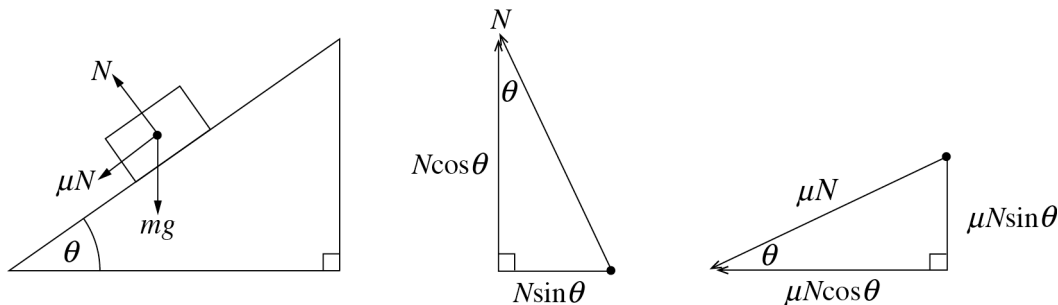
adding $\alpha^4 + \beta^4 + \gamma^4 = p(\alpha^2 + \beta^2 + \gamma^2) - q(\alpha + \beta + \gamma)$

$$= 8 \times 16 - q \times 0$$

$$= 128$$

Question 14 (c) (i)

Criteria	Marks
• Provides correct solution	3
• Provides correct resolution of forces, or equivalent merit	2
• Resolves forces in vertical direction, or equivalent merit	1

Sample answer:

Resolving forces:

Vertically $N \cos \theta - \mu N \sin \theta = mg$ (1)

Horizontally $N \sin \theta + \mu N \cos \theta = \frac{mv^2}{r}$ (2)

Dividing (2) by (1)

$$\frac{\sin \theta + \mu \cos \theta}{\cos \theta - \mu \sin \theta} = \frac{v^2}{rg}$$

$$\therefore v^2 = rg \left(\frac{\sin \theta + \mu \cos \theta}{\cos \theta - \mu \sin \theta} \right)$$

Dividing by $\frac{\cos \theta}{\cos \theta}$:

$$v^2 = rg \left(\frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right)$$

Question 14 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Obtains correct expression for $\tan \theta$ in terms of μ , or equivalent merit	1

Sample answer:Since $v = V$

$$V^2 = rg \text{ and } V^2 = rg \left(\frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right)$$

$$\text{so } \frac{\tan \theta + \mu}{1 - \mu \tan \theta} = 1$$

hence

$$\tan \theta + \mu = 1 - \mu \tan \theta$$

$$\mu + \mu \tan \theta = 1 - \tan \theta$$

$$\text{so } \mu = \frac{1 - \tan \theta}{1 + \tan \theta}$$

$$\text{since } 0 < \theta < \frac{\pi}{2}, \tan \theta > 0$$

$$\text{so } 1 - \tan \theta < 1 \text{ and } 1 + \tan \theta > 1$$

$$\therefore \mu < 1$$

Question 15 (a) (i)

Criteria	Marks
• Provides correct solution	2
• Obtains $\ddot{x} = -kv^2$, or equivalent merit	1

Sample answer:

$$0 \quad \xrightarrow{\quad \quad \quad} \quad \quad \quad \longleftarrow -kv^2$$

$$\ddot{x} = -kv^2$$

$$\frac{dv}{dt} = -kv^2$$

$$k dt = -\frac{dv}{v^2}$$

$$\Rightarrow kt = -\int \frac{dv}{v^2}$$

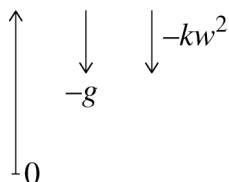
$$kt = \frac{1}{v} + C$$

$$\text{Now } t = 0, v = u \Rightarrow C = -\frac{1}{u}$$

$$\therefore \frac{1}{v} = kt + \frac{1}{u}$$

Question 15 (a) (ii)

Criteria	Marks
• Provides correct solution	3
• Correctly integrates to find t , or equivalent merit	2
• Obtains $\ddot{x} = -g - kw^2$, or equivalent merit	1

Sample answer:

$$\ddot{x} = -g - kw^2$$

$$\therefore \frac{dw}{dt} = -(g + kw^2)$$

$$\frac{dt}{dw} = -\frac{1}{g + kw^2}$$

$$t = -\int \frac{1}{g + kw^2} dw$$

$$= -\frac{1}{k} \int \frac{1}{\frac{g}{k} + w^2} dw$$

$$= -\frac{1}{k} \left[\sqrt{\frac{k}{g}} \tan^{-1} \left(w \sqrt{\frac{k}{g}} \right) \right] + C$$

$$\therefore t = -\frac{1}{\sqrt{gk}} \tan^{-1} \left(w \sqrt{\frac{k}{g}} \right) + C$$

Now $t = 0$, $w = u$, so

$$C = \frac{1}{\sqrt{gk}} \tan^{-1} \left(u \sqrt{\frac{k}{g}} \right)$$

$$\therefore t = \frac{1}{\sqrt{gk}} \left(\tan^{-1} \left(u \sqrt{\frac{k}{g}} \right) - \tan^{-1} \left(w \sqrt{\frac{k}{g}} \right) \right)$$

Question 15 (a) (iii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

The second particle is at rest when $w = 0$

$$\therefore t = \frac{1}{\sqrt{gk}} \tan^{-1} \left(u \sqrt{\frac{k}{g}} \right)$$

At this time the velocity of the first particle is

$$\frac{1}{V} = \frac{1}{u} + k \left(\frac{1}{\sqrt{gk}} \tan^{-1} \left(u \sqrt{\frac{k}{g}} \right) \right) \text{ from (i)}$$

$$\therefore \frac{1}{V} = \frac{1}{u} + \sqrt{\frac{k}{g}} \tan^{-1} \left(u \sqrt{\frac{k}{g}} \right)$$

Question 15 (a) (iv)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$\text{From (iii), } \frac{1}{V} = \frac{1}{u} + \sqrt{\frac{k}{g}} \tan^{-1} \left(u \sqrt{\frac{k}{g}} \right)$$

$$\text{for } u \text{ range, } \frac{1}{u} \approx 0, \tan^{-1} \left(u \sqrt{\frac{k}{g}} \right) \approx \frac{\pi}{2}$$

$$\therefore \frac{1}{V} \approx \sqrt{\frac{k}{g}} \cdot \frac{\pi}{2} \Rightarrow V \approx \frac{2}{\pi} \sqrt{\frac{g}{k}}$$

Question 15 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Obtains one of the required inequalities	1

Sample answer:Since for $x \geq 0$

$$1 - x^2 \leq 1 \quad \text{ie } (1-x)(1+x) \leq 1$$

since $(1+x) > 0$ we have $1-x \leq \frac{1}{1+x}$

and since $x \geq 0$ then $1+x \geq 1$

$$\therefore \frac{1}{1+x} \leq 1$$

$$\therefore 1-x \leq \frac{1}{1+x} \leq 1$$

Question 15 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Integrates the inequalities from part (i), or equivalent merit	1

Sample answer:

$$\begin{aligned}\text{Since } \int_a^b \frac{1}{1+x} dx &= \left[\ln(1+x) \right]_a^b \\ &= \ln(1+b) - \ln(1+a)\end{aligned}$$

We want $\ln\left(1 + \frac{1}{n}\right)$ so let $a = 0$ and $b = \frac{1}{n}$

$$\therefore \int_0^{\frac{1}{n}} 1-x \, dx \leq \int_0^{\frac{1}{n}} \frac{1}{1+x} \, dx \leq \int_0^{\frac{1}{n}} 1 \, dx$$

$$\therefore \left[x - \frac{x^2}{2} \right]_0^{\frac{1}{n}} \leq \left[\ln(1+x) \right]_0^{\frac{1}{n}} \leq \left[x \right]_0^{\frac{1}{n}}$$

$$\therefore \frac{1}{n} - \frac{1}{2n^2} \leq \ln\left(1 + \frac{1}{n}\right) \leq \frac{1}{n}$$

multiplying through by n :

$$1 - \frac{1}{2n} \leq n \ln\left(1 + \frac{1}{n}\right) \leq 1$$

Question 15 (b) (iii)

Criteria	Marks
• Provides correct explanation	1

Sample answer:

$$\text{From (ii), } 1 - \frac{1}{2n} \leq n \ln \left(1 + \frac{1}{n} \right) \leq 1$$

$$\Rightarrow 1 - \frac{1}{2n} \leq \ln \left(1 + \frac{1}{n} \right)^n \leq 1$$

$$\lim_{n \rightarrow \infty} \left(1 - \frac{1}{2n} \right) \leq \lim_{n \rightarrow \infty} \ln \left(1 + \frac{1}{n} \right)^n \leq \lim_{n \rightarrow \infty} 1$$

$$\text{ie } 1 \leq \lim_{n \rightarrow \infty} \ln \left(1 + \frac{1}{n} \right)^n \leq 1$$

$$\Rightarrow \lim_{n \rightarrow \infty} \ln \left(1 + \frac{1}{n} \right)^n = 1$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e$$

Question 15 (c) (i)

Criteria	Marks
• Provides correct proof	1

Sample answer:

Replacing x with x^2 , y with y^2 in the given inequality we have

$$\sqrt{x^2 y^2} \leq \frac{x^2 + y^2}{2}$$

$$\text{ie } xy \leq \frac{x^2 + y^2}{2}$$

$$\text{so } \sqrt{xy} \leq \sqrt{\frac{x^2 + y^2}{2}}, \text{ since } x, y > 0$$

Question 15 (c) (ii)

Criteria	Marks
• Provides correct proof	2
• Uses (i) to obtain a valid expression in a^2 , b^2 , c^2 and d^2 , or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \sqrt{(ab)(cd)} &= \sqrt{ab}\sqrt{cd} \\
 &\leq \sqrt{\frac{a^2+b^2}{2}}\sqrt{\frac{c^2+d^2}{2}} \quad \text{by (i)} \\
 &\leq \sqrt{\left(\frac{a^2+b^2}{2}\right)\left(\frac{c^2+d^2}{2}\right)} \\
 &\leq \frac{1}{2}\left(\frac{a^2+b^2+c^2+d^2}{2}\right) \quad \text{by given inequality} \\
 &= \frac{a^2+b^2+c^2+d^2}{4}
 \end{aligned}$$

Taking positive square roots,

$$\sqrt[4]{abcd} \leq \sqrt{\frac{a^2+b^2+c^2+d^2}{4}}$$

Question 16 (a) (i)

Criteria	Marks
• Provides correct solution	2
• Shows total number of arrangements is $\binom{15}{5}$, or equivalent merit	1

Sample answer:

There are $\binom{15}{5}$ ways to place 5 black counters in the grid.

There are three choices to place a black counter in each column, so there are 3^5 ways to place the black counters with one in each column.

$$\therefore \text{Probability} = \frac{3^5}{\binom{15}{5}} = \frac{81}{1001}$$

Question 16 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Shows total number of arrangements is $\binom{nq}{q}$, or equivalent merit	1

Sample answer:

There are nq places and we choose q places for black counters, giving $\binom{nq}{q}$ arrangements.

There are n places for each black counter in each of q columns.

So there are n^q arrangements with one black counter in each column.

$$P_n = \frac{n^q}{\binom{nq}{q}}$$

Question 16 (a) (iii)

Criteria	Marks
• Provides correct solution	2
• Obtains an expression for P_n with q terms in denominator, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 P_n &= \frac{n^q}{\binom{nq}{q}} \\
 &= \frac{n^q}{nq(nq-1) \dots (nq-q+1)/q!} \\
 &= \frac{q!n^q}{nq(nq-1) \dots (nq-q+1)} \\
 &= \frac{q!}{q\left(q-\frac{1}{n}\right)\left(q-\frac{2}{n}\right) \dots \left(q-\frac{q}{n}+\frac{1}{n}\right)}
 \end{aligned}$$

As $n \rightarrow \infty$,

$$P_n \rightarrow \frac{q!}{q^q}$$

Question 16 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Applies De Moivre's theorem, or equivalent merit	1

Sample answer:

By De Moivre's theorem

$$(\cos \alpha + i \sin \alpha)^{2n} = \cos(2n\alpha) + i \sin(2n\alpha)$$

By the binomial theorem,

$$\begin{aligned} (\cos \alpha + i \sin \alpha)^{2n} &= \cos^{2n} \alpha + \binom{2n}{1} \cos^{2n-1} \alpha \cdot i \sin \alpha + \binom{2n}{2} \cos^{2n-2} \alpha (i \sin \alpha)^2 + \dots \\ &\quad + \binom{2n}{2n-2} \cos^2 \alpha (i \sin \alpha)^{2n-2} + \binom{2n}{2n-1} \cos \alpha (i \sin \alpha)^{2n-1} + \binom{2n}{2n} (i \sin \alpha)^{2n} \end{aligned}$$

Equating real parts,

$$\cos(2n\alpha) = \cos^{2n} \alpha - \binom{2n}{2} \cos^{2n-2} \alpha \sin^2 \alpha + \dots + \binom{2n}{2n-2} \cos^2 \alpha \sin^{2n-2} \alpha (-1)^{n-1} + (-1)^n \sin^{2n} \alpha$$

Question 16 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Explains the connection between $T_{2n}(x)$ and $\cos(2n\alpha)$, or equivalent merit	1

Sample answer:

$$T_{2n}(x) = \cos(2n \cos^{-1} x)$$

put $\alpha = \cos^{-1} x$, so $x = \cos \alpha$ in (i)noting that $\sin^{2k} \alpha = (\sin^2 \alpha)^k$

$$= (1 - \cos^2 \alpha)^k$$

Then

$$T_{2n}(x) = x^{2n} - \binom{2n}{2} x^{2n-2} (1-x^2) + \dots + (-1)^n (1-x^2)^n$$

Question 16 (b) (iii)

Criteria	Marks
• Provides correct solution	3
• Shows that the roots of $T_{2n}(x)$ are $\cos\left(\frac{\pi}{4n}\right), \cos\left(\frac{3\pi}{4n}\right), \dots, \cos\left(\frac{(4n-1)\pi}{4n}\right)$, or equivalent merit	2
• Obtains (or verifies) that $\cos\left(\frac{(2k-1)\pi}{4n}\right)$ are all roots of $T_{2n}(x)$, or equivalent merit	1

Sample answer:

$$0 = T_{2n}(x) = \cos(2n \cos^{-1} x)$$

$$\text{This occurs when } 2n \cos^{-1} x = \frac{(2k-1)\pi}{2}$$

$$\text{ie } \cos^{-1} x = \frac{(2k-1)\pi}{4n}$$

$$x = \cos\left(\frac{(2k-1)\pi}{4n}\right)$$

Consider the $2n$ solutions with $k = 1, 2, \dots, 2n$ ie

$$\cos\left(\frac{\pi}{4n}\right), \cos\left(\frac{3\pi}{4n}\right), \dots, \cos\left(\frac{(4n-1)\pi}{4n}\right)$$

As the angles are in the range 0 to π and are distinct these solutions are distinct. (Given that $\cos x$ is decreasing for $0 \leq x \leq \pi$.)

We have $2n$ distinct solutions of a polynomial of degree $2n$, and so these are all the roots.

As we have a polynomial of even degree, the product of the roots is the constant term divided by the leading term. We may find the constant term by setting $x = 0$, obtaining $(-1)^n$.

The coefficients of x^{2n} are

$$1, \quad \binom{2n}{2}, \quad \binom{2n}{4}, \quad \dots, \quad 1$$

The leading term is the sum of these and hence

$$\cos\left(\frac{\pi}{4n}\right)\cos\left(\frac{3\pi}{4n}\right)\cos\left(\frac{(4n-1)\pi}{4n}\right) = \frac{(-1)^n}{1 + \binom{2n}{2} + \binom{2n}{4} + \dots + 1}$$

In fact the denominator on the right may be simplified

$$\cos\left(\frac{\pi}{4n}\right)\cos\left(\frac{3\pi}{4n}\right)\cos\left(\frac{(4n-1)\pi}{4n}\right) = \frac{(-1)^n}{2^{2n-1}}$$

Question 16 (b) (iv)

Criteria	Marks
• Provides correct proof	2
• Evaluates $T_{2n}\left(\frac{1}{\sqrt{2}}\right)$, or equivalent merit	1

Sample answer:

Put $x = \frac{1}{\sqrt{2}}$ then $1 - x^2 = \frac{1}{2}$

$$T_{2n}\left(\frac{1}{\sqrt{2}}\right) = \frac{1}{2^n} - \binom{2n}{2} \cdot \frac{1}{2^n} + \binom{2n}{4} \cdot \frac{1}{2^n} - \dots + (-1)^n \binom{2n}{2n} \cdot \frac{1}{2^n}$$

$$= \cos\left(2n \cos^{-1} \frac{1}{\sqrt{2}}\right) = \cos\left(\frac{n\pi}{2}\right)$$

$$\therefore 1 - \binom{2n}{2} + \binom{2n}{4} - \binom{2n}{6} + \dots + (-1)^n \binom{2n}{2n} = 2^n \cos\left(\frac{n\pi}{2}\right)$$

2015 HSC Mathematics Extension 2

Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	3.2	E3
2	1	2.1	E3
3	1	1.4	E6
4	1	7.2	E4
5	1	2.4	E3
6	1	4.1	E8
7	1	8	HE3, E9
8	1	1.7	E6
9	1	2.2	E3
10	1	8	E2

Section II

Question	Marks	Content	Syllabus outcomes
11 (a)	2	2.1	E3
11 (b) (i)	1	2.2	E3
11 (b) (ii)	1	2.2	E3
11 (b) (iii)	1	2.2	E3
11 (c)	2	7.6	E4
11 (d)	2	3.1	E3
11 (e)	3	1.8	E6
11 (f) (i)	2	4.1	PE6, E2
11 (f) (ii)	1	4.1	E8
12 (a) (i)	1	2.2	E3
12 (a) (ii)	1	2.2	E3
12 (a) (iii)	1	2.3	E3
12 (b) (i)	3	7.4, 7.5	E3, E4
12 (b) (ii)	1	7.4, 7.5	E3, E4
12 (c) (i)	2	1.8	E6
12 (c) (ii)	2	1.8	E6
12 (d)	4	5.1	E7
13 (a) (i)	1	3.2	E3
13 (a) (ii)	2	3.2	E3
13 (a) (iii)	3	3.2	E3
13 (b) (i)	1	8	E2

Question	Marks	Content	Syllabus outcomes
13 (b) (ii)	2	5.1	E7
13 (c) (i)	2	14.1E, 8	HE3, E8
13 (c) (ii)	2	14.1E, 8	HE3, E8
13 (c) (iii)	2	14.1E, 8, 4.1	HE3, E8
14 (a) (i)	2	4.1	E8
14 (a) (ii)	2	4.1	E8
14 (a) (iii)	1	4.1	E8
14 (b) (i)	1	7.5	E4
14 (b) (ii)	2	7.5	E4
14 (b) (iii)	2	7.5	E4
14 (c) (i)	3	6.3.4	E5
14 (c) (ii)	2	6.3.4	E5
15 (a) (i)	2	6.2.1	E5
15 (a) (ii)	3	6.2.2	E5
15 (a) (iii)	1	6.2.1, 6.2.2	E5
15 (a) (iv)	1	6.2.1	E5
15 (b) (i)	2	8, 3, 1.4E	E2, HE7
15 (b) (ii)	2	8, 1.4E, 12.5	PE3, HE7, E2
15 (b) (iii)	1	8, 1.4E	PE3, E2
15 (c) (i)	1	8.3	E2
15 (c) (ii)	2	8.3	E2
16 (a) (i)	2	8	E2
16 (a) (ii)	2	8	E2
16 (a) (iii)	2	8	E2
16 (b) (i)	2	2.4	E3
16 (b) (ii)	2	8	E4
16 (b) (iii)	3	7.5	E4
16 (b) (iv)	2	7.5	E9